

Project Description:

Sundial flow-matches from Gaussian white noise with a small adaptive-layer-norm MLP as the velocity net. TSFlow flow-matches from a Gaussian-process prior with a DiffWave stack whose residual blocks are structured state-space layers (S4). Those two choices are usually discussed as *priors*. This project treats them as a *head* question.

The narrow cell: independent-coupling conditional flow matching in raw observation coordinates ($\mathbb{R}^F$), source and target both smooth (same or matched kernel), velocity net an unstructured AdaLN MLP. Claim: that regression is poorly preconditioned — common-mode gradients dominate, residual coordinates jitter — whereas the same pair is native food for an S4. Conditioning on the lookback hidden state still identifies the destination; the pairing arrows do not cancel. What is at issue is the Jacobian of the head, not identifiability.

This is not “Sundial fails on smooth data.” Sundial never trains with a Gaussian-process source. It is not “flow matching from a GP fails.” TSFlow already does that, with a different head. Chronos-Bolt’s residual quantile readout is out of scope: it never sees a source patch.

The project does not pretrain on TimeBench, does not freeze a Sundial checkpoint, and does not replace the prior in [ideas/latent-gauss-markov-prior.md](#). It trains tiny velocity nets on synthetic patches and writes a design constraint: whether a structured observation-space source may sit under TimeFlow’s head, or needs a temporally structured (or whitened) map. Every cell is scored against a pre-written alternative.

Project Tasks:

Task 1: Landscape and the synthetic pair (Minimum Passing Score)

Write a short critical review that places the cell against (i) Sundial’s TimeFlow head (AdaLN MLP, white-noise source, multi-patch horizon), (ii) TSFlow’s velocity net (DiffWave residual blocks, S4 along time, Gaussian-process source, often lookback-conditioned). State explicitly that TSFlow is not a counterexample until the heads match, and that white-noise $\rightarrow$ smooth target is Sundial’s published geometry, not the stress test.

Then build the data. Sample univariate lookback and horizon patches from one Gaussian process with a long lengthscale (near-collinear (F)-vectors after the same instance norm Sundial uses). Fix (F) in advance (e.g. 16). Two source families, drawn independently of the target under the usual conditional-flow-matching coupling:

- isotropic Gaussian white noise in $\mathbb{R}^F$;
- a Gaussian-process draw with the **same** kernel as the data (matched smooth source).

Record the spectrum of the target patches and of the velocity ($y\text{-}\epsilon$) under each source: fraction of variance in the first two principal components. If that fraction is not high for the matched-GP velocity (e.g. > 0.9), the lengthscale is too short and Task 2 must not run.

Measurable completion. A one-page related-work note that cites Sundial, TSFlow, and S4 and names the empty cell; plus a table of principal-component energy for targets and for both velocity families, on a held-out synthetic draw.

Conclusions that still count if the numbers go either way.

- Matched-GP velocity is near rank-1/2, white-noise velocity is not: the numerical contrast exists; proceed.
- Both velocities are full rank: the kernel is not smooth on this (F); lengthen the lengthscale or shorten the patch, do not interpret Task 2.
- Both velocities are collapsed: the series are constants after instance norm; there is no task (see Task 2, empty-task reading).

Task 2: The 2×2 — source × head (Average Score)

Train two velocity nets of comparable width, both conditioned on a *tiny* lookback encoder (the same encoder for both heads; not a foundation backbone):

- Sundial-style AdaLN MLP: interpolant ($y^{\{t\}}$), flow time (t), lookback vector (h), output in $(\mathbb{R}^F)$;
- TSFlow-style stack: three residual blocks, each with an S4 layer along the patch, same condition (h).

Cross with the two sources from Task 1. Four cells. Evaluate on held-out synthetic windows, **not** with horizon mean-squared error alone:

1. **Common-mode velocity error.** Mean-squared error of the velocity projected onto the leading principal component of ($y - \text{varepsilon}$).
2. **Residual-PC velocity error.** The same, on the orthogonal complement (principal components (3, ..., F), or whatever Task 1 left outside the first two).
3. **Sample-path roughness.** After Euler integration for a fixed (K), mean adjacent jump ($|x_{i+1} - x_i|$) of generated patches, versus the same quantity on the GP targets.

Declare a head “works” in a cell only if residual-PC error is within (20%) of common-mode error *and* generated roughness is within (20%) of the target roughness. Common-mode error alone is not enough.

Measurable completion. A 2×2 table of the three numbers, plus one plot of generated patches versus targets in each cell. No TimeBench, no pretrained Sundial weights.

Conclusions that still count if the numbers go either way.

- **Head, not prior.** AdaLN MLP matches the common mode and loses on residual principal components / roughness in the GP→GP cell; S4 does not. A structured source in observation space needs a temporally structured velocity net. That is a design constraint on [ideas/latent-gauss-markov-prior.md](#).
- **Prior was overstated.** AdaLN MLP matches S4 on all three numbers in the GP→GP cell. The rank-deficiency story is wrong; TimeFlow’s head may sit under a Gauss–Markov source.
- **S4 was doing the GP’s job.** S4 wins even from white noise; the matched-GP source adds nothing once the head has a temporal law. Then the lookback-plus-S4 already supplies the smoothness (TSFlow’s conditional moral).
- **The source is the knob.** Same smooth targets: white-noise source works for the AdaLN MLP, matched-GP source does not (or the reverse). Isolated the source; Sundial-as-published is off the hook.
- **Empty task.** All four cells fail the “works” rule, including S4. Shorten the lengthscale or lengthen the patch and rerun Task 1; do not write a method conclusion.

Task 3: A third head, and the recommendation (Maximum Score)

Keep the Task 2 encoder and the matched-GP source. Add one control head that *removes* the unused coordinates by design: PCA-whiten the patch (fit on training velocities) and run the same AdaLN MLP in that basis, or replace the readout by a degree-1 polynomial / first-two-PC map. Score it with the same three numbers.

Optionally, repeat the GP→GP column of Task 2 on **one** real family that is already smooth after instance norm (a long-lengthscale GIFT-Eval split, or a sine-plus-slow-trend). If the real series is not near-collinear by the Task 1 spectrum test, skip it and say so; do not expand to a benchmark.

Write a one-paragraph recommendation for the latent-prior programme: keep TimeFlow's head, switch to S4, or flow-match only after whitening / in a latent with a temporal law.

Measurable completion. The third-head row on the synthetic GP→GP cell (three numbers); the recommendation paragraph; the optional real-series column only if the spectrum test passes.

Conclusions that still count if the numbers go either way.

- Whitening or the polynomial head closes the AdaLN gap: the extra (F-2) channels were unused; the fix is a change of coordinates, not a bigger Transformer.
- The third head does *not* close the gap, and S4 still wins: the failure is the lack of a *temporal* law in the velocity net, not only collinearity. Then a latent Gauss–Markov source still wants an S4-like (or SSM-like) head, not a whitened MLP.
- The third head *and* S4 tie the raw AdaLN MLP: Task 2's "head" reading was a capacity or optimisation artefact; report that and keep Sundial's head in the larger proposal.
- Real series disagree with the synthetic GP→GP cell: the stress test is real but not the geometry of that dataset; do not generalise past the synthetic pair.

A fitted velocity net is a *description* of this synthetic cell. The project does not claim that Sundial *is* ill-conditioned on TimeBench.

Project Completeness: 1 - 30% 2 - 40% 3 - 30%

References:

- [1] Liu, Y., et al. "Sundial: A Family of Highly Capable Time Series Foundation Models." ICML, 2025. <https://arxiv.org/abs/2502.00816>
- [2] Kollovieh, M., et al. "Flow Matching with Gaussian Process Priors for Probabilistic Time Series Forecasting." ICLR, 2025. <https://arxiv.org/abs/2410.03024>
- [3] Gu, A., Goel, K., and Ré, C. "Efficiently Modeling Long Sequences with Structured State Spaces." ICLR, 2022. <https://arxiv.org/abs/2111.00396>
- [4] Lipman, Y., et al. "Flow Matching for Generative Modeling." ICLR, 2023. <https://arxiv.org/abs/2210.02747>
- [5] Peebles, W., and Xie, S. "Scalable Diffusion Models with Transformers." ICCV, 2023. <https://arxiv.org/abs/2212.09748>
- [6] Kong, Z., et al. "DiffWave: A Versatile Diffusion Model for Audio Synthesis." ICLR, 2021. <https://arxiv.org/abs/2009.09761>
- [7] Gu, A., et al. "HiPPO: Recurrent Memory with Optimal Polynomial Projections." NeurIPS, 2020. <https://arxiv.org/abs/2008.07669>

[8] Aksu, T., et al. "GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation." 2024.
<https://arxiv.org/abs/2410.10393>